# Compressing dense similarity for dataset splitting with global and local control

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**Abstract**

Similarity-aware evaluation requires separating related entities across data folds, but their total overlap is a dense pairwise quantity. We introduce low-rank similarity splitting, which uses a Nyström factor to compress the kernel into shared coordinates and optimizes cross-fold overlap through their fold sums. At fixed rank, factor storage grows linearly with dataset size, and capacity constraints control the returned fold counts. On six molecular datasets, the largest measured rank-256 cut error for the ordinary kernel is 0.0058 of its off-diagonal similarity mass. The plain method partitions two million molecular fingerprints in 19.9 seconds on an H100 NVL with a 1% fold-size corridor. Kernel powers and capped groups add control over close-analogue contamination, allowing a trade-off between global and local overlap. Comparisons with DataSAIL and an ordinary graph partitioner show dataset-dependent quality–cost trade-offs. Independent fingerprints and ten external bioactivity assays establish the scope of this control: separation depends on the representation, and strong chemical-series holdouts can induce target shift and chance-level prediction. Low rank provides a compact framework for constructing capacity-controlled generalization tests whose scientific purpose can be assessed explicitly.

**Keywords:** dataset splitting; molecular machine learning; Nyström approximation; information leakage; scientific model evaluation.

## 1 Introduction

An evaluation split defines the generalization task a model must solve. In molecular machine learning, predicting another member of a known chemical series differs from predicting a new series. Random, scaffold, cluster and temporal splits impose different boundaries between these tasks and can change apparent model performance [Wu et al., 2018, Yang et al., 2019]. Lo-Hi makes the distinction between lead optimization and novel-hit identification explicit [Steshin, 2023]. Temporal studies and SIMPD further show that the appropriate degree of separation depends on the intended project scenario [Sheridan, 2013, Landrum et al., 2023]. Evaluation guidelines and activity-cliff benchmarks also motivate examining response variation alongside structural similarity [Bender et al., 2022, van Tilborg et al., 2022].

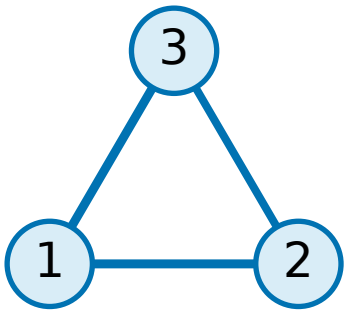
Similarity offers a direct way to specify such a boundary: minimize the total similarity crossing folds while preserving their requested sizes. This quantity includes every pair of entities. Storing it explicitly requires quadratic memory, making repeated split construction

costly as datasets grow. DataSAIL addresses the optimization through clustering and integer programming [Joeres et al., 2025, 2026]. A sparse nearest-neighbour graph offers another representation, retaining strong local connections while omitting most pairs. These choices raise a computational question: can we optimize an approximation to the dense overlap without storing the dense matrix?

We address this question by compressing the kernel into a low-rank factor. Each entity receives a short vector, and the dot product of two vectors approximates their similarity. Crucially, cross-fold overlap can then be computed from the sum of vectors in each fold. The solver works with these compact sums and specified integer capacities, rather than an explicit pairwise matrix. A sparse graph represents a selected set of pairs; the factor instead approximates similarities through coordinates shared by all entities (Figure 1). This is useful when similarity contains redundancy that a modest rank captures. Nyström approximation and its connections to kernel clustering are established [Williams and Seeger, 2000, Dhillon et al., 2004, Gittens and Mahoney, 2016, Wang et al., 2019, Chitta et al., 2011]; our contribution is a capacity-constrained splitting framework and an evaluation of when its compressed objective is useful.

Making the dense objective affordable also exposes a distinction in what it measures. A sum over many pairs can assign little importance to a few close analogues crossing the split, although those analogues may strongly affect prediction. We therefore measure total overlap and nearest-training-neighbour similarity separately. Integer kernel powers emphasize strong similarities, while capped groups encourage nearby entities to stay together. Both operate within the factor framework, allowing explicit choices between global overlap, local contamination and fold capacities. The powered kernel has its own approximation requirements, and grouping adds neighbour-construction work.

The study asks three linked questions: how well compression preserves cut values, how the returned partitions control global and local overlap relative to established alternatives, and what that separation means for prediction. We test fidelity and rank sensitivity, compare full-size molecular and materials benchmarks, and measure computational scaling separately for the plain and combined settings. Synthetic families and per-molecule errors test the meaning of the local metric. External assay tasks then examine the consequences of fixed settings, including target-composition shifts. Together these experiments establish a practical use for low rank: constructing and comparing specified similarity-aware generalization tests at controlled fold sizes.




**Coarsen**
Weighted supernodes
and inter-group edges



**Sparsify**

Select nearby pairs



**Compress**
Shared coordinates
approximate all pairs

Fold sum p(c) = Σ B(i) for i in fold c
Cut = ½ [ ‖Σ p(c)‖² − Σ ‖p(c)‖² ]

Figure 1: Three approaches to representing similarity for splitting. The schematic illustrates coarsening into weighted supernodes with inter-group connections, selecting nearby pairwise edges, and representing similarities through shared low-rank coordinates. Coarsening can retain inter-cluster similarity; it does not imply removing every connection between clusters. Compression supplies the block-sum cut identity in Equation 2; power and grouping then control the global–local trade-off. The matrices are illustrative.

# 2 Low-rank similarity splitting

## 2.1 Problem and evaluation quantities

Let $S \in \mathbb{R}^{n \times n}$ be a symmetric, nonnegative, positive-semidefinite kernel over $n$ entities and let $z_i \in \{1, \ldots, k\}$ denote the fold of entity $i$. For a partition $z$, define the unordered cross-fold similarity

$$\mathcal{C}_S(z) = \sum_{i<j} S_{ij}\, \mathbf{1}[z_i \neq z_j], \qquad L_S(z) = \frac{\mathcal{C}_S(z)}{\sum_{i<j} S_{ij}}. \tag{1}$$

The denominator is independent of the partition and the diagonal contributes nothing. We use $L_S$ only when the off-diagonal total is positive. Signed cosine kernels can be positive semidefinite while violating nonnegativity; their ratios do not have the same interpretation and are outside our leakage claims.

We evaluate global overlap relative to the random expectation at the realized fold sizes. A separate local score measures each test entity's similarity to its nearest training neighbour. For molecular fingerprints, a ramp from 0.4 to 0.8 emphasizes close analogues. Section 6 defines the scoring and count-preserving random normalization precisely.

## 2.2 Factor-space formulation

Sample $r$ landmark entities, form the entity–landmark matrix $C$ and landmark kernel $W$, and construct $B = CW^{-1/2}$ using a regularized eigendecomposition. This produces a regularized Nyström approximation $\widetilde{S} = BB^\top$. For the sum of factor rows in fold $c$, $p_c = \sum_{i:z_i=c} B_i$, the approximate cut satisfies

$$\mathcal{C}_{\widetilde{S}}(z) = \frac{1}{2}\left(\left\|\sum_c p_c\right\|^2 - \sum_c \|p_c\|^2\right). \tag{2}$$

Equation 2 is exact for $\widetilde{S}$, and computing the block sums requires $O(nr)$ work. It is an approximation to the cut of $S$, not a certificate that the original cut has been minimized. Supplementary Information gives a uniform objective-error bound and separates it from optimization error.

## 2.3 Capacities and optimization

Largest-remainder rounding converts requested ratios into integer fold targets summing to $n$. Zero size tolerance enforces those counts exactly; positive tolerance permits a two-sided corridor containing them. Each restart alternates between computing fold sums and proposing capacity-constrained assignments. A proposal is accepted when it does not increase the approximate cut, and the best restart undergoes greedy move or swap refinement [Fiduccia and Mattheyses, 1982, Kernighan and Lin, 1970]. At exact capacities, swaps preserve fold counts. For a move from fold $a$ to $b$, the change in cut is

$$\Delta\mathcal{C} = B_i^\top p_a - B_i^\top p_b - \|B_i\|^2. \tag{3}$$

Thus refinement also uses factor rows and fold sums. Finite restarts and a restricted swap search provide a heuristic solution. The standard configuration uses rank 256, four restarts and up to 25 assignment iterations. The supplement specifies assignment, landmark selection and stopping rules. Table 1 summarizes the workflow.

## 2.4 Kernel sharpening and capped groups

For an integer $p \geq 1$, we factorize the elementwise power $S^{\circ p}$. The Schur product theorem preserves positive semidefiniteness. Powering reduces weak background similarities relative to strong ones: $0.1^3 = 0.001$, whereas $0.7^3 = 0.343$. More formally, if $N_\tau(z)$ counts crossing pairs with similarity at least $\tau > 0$, then $\mathcal{C}_{S^{\circ p}}(z) \geq \tau^p N_\tau(z)$ (Supplementary Information). The powered cut therefore penalizes strong crossing pairs, although it differs from the average nearest-neighbour evaluation score. All leakage comparisons use the same original kernel, regardless of the optimization power.

Grouping complements this change of weighting by restricting assignments. We process top-32-neighbour edges above a threshold $\tau$ in descending order, merging components subject to a count cap. A group is represented by the sum of its factor rows and its entity count, so the same cut and move identities apply to whole-group assignments. The combined molecular setting uses $p = 3$, $\tau = 0.4$ and a cap of half the smallest target fold. For non-Tanimoto features, $\tau$ is the 0.995 pair-similarity quantile. The builder stops component growth after a cap-blocked merge. If whole-group repair fails to satisfy capacities, it can dissolve groups into singletons; we report this relaxation and check the returned counts. Grouping consequently encourages local separation while retaining all valid input entities. Omitted neighbour edges, capped merges and repair prevent a universal threshold-exclusion guarantee.

| | Low-rank similarity splitting |
|---|---|
| 1 | Round requested capacities and check that every target fold is nonempty. |
| 2 | Select landmarks and build a rank-$r$ factor of $S^{\circ p}$. |
| 3 | If grouping is enabled, construct capped neighbour groups and sum their factor rows; retain their entity counts as weights. |
| 4 | For each seeded restart, alternate fold sums with capacity-constrained assignment proposals. Repair weighted capacities, recording dissolved groups. |
| 5 | Retain the best restart and refine through legal moves or count-preserving swaps. Accept operations that decrease the factor cut. |
| 6 | Verify final capacities and return every entity's fold with diagnostics. |

Table 1: Algorithm outline. Fold sums replace repeated dense-cut evaluation. All acceptance tests concern the approximate kernel. The supplement specifies the proposal, repair and stopping rules.

## 2.5 Computational scope

At rank $r$, factor storage and fold-sum construction require $O(nr)$ space and work, respectively. This compact representation makes the dense objective accessible at large $n$. Total

runtime also depends on factor construction, assignment and refinement. The combined setting adds neighbour construction, with quadratic arithmetic for exact similarities. We measure its scaling separately from the plain method; the supplement gives the operation-by-operation costs, stopping rules and neighbour-backend details.

# 3 Results

## 3.1 Compression preserves the ordinary dense cut at modest rank

The first question is whether a compact factor represents the cut values that the solver is intended to improve. Across six full development datasets, three factor seeds and five partition perturbations per factor, the largest rank-256 error for the ordinary kernel is 0.0058 of its off-diagonal similarity mass. Error decreases with rank (Figure 2). Thus, in this collection, a modest factor supports accurate evaluation of the dense cut without storing its full matrix. This tests objective fidelity across the examined partitions; optimization quality remains a separate question.

For comparison, the top-32 graph retains a mean 10.6% of the ordinary kernel's similarity mass on the same six datasets. Raising its degree to 128 increases this to 28.2%. Compression and sparsification thus retain different aspects of the dense objective. The graph-degree study and its returned-partition scores appear in the supplement.

## 3.2 Local similarity measures a distinct aspect of separation

A faithful dense objective is useful only if its scientific meaning is clear. On synthetic datasets with known families, local leakage correlates with cross-fold family contamination at Spearman $\rho = 0.519$, compared with 0.020 for the global ratio on the same 120 observations. Pooled correlations span several constructed regimes; their individual results appear in the supplement. The continuum regime favors the global score, whereas four other regimes favor the local score. This supports measuring close-neighbour contamination separately from total overlap.

Predictions provide a second interpretation. In the random and plain-low-rank D-MPNN partitions, similarity to the nearest training molecule correlates negatively with absolute ensemble error on ESOL, FreeSolv and Lipophilicity ($\rho = -0.307$, $-0.407$ and $-0.281$, respectively). On Lipophilicity, the per-point global score has a weak positive correlation ($+0.049$). BBBP is an exception to the local trend ($\rho = +0.052$), and within-split associations vary. Local similarity therefore captures an empirically relevant aspect of prediction, with a model- and dataset-dependent relationship to error. Within-split analyses and prediction provenance are supplied in the supplement.

## 3.3 Power and grouping control local overlap, with a fidelity cost

The second question is whether the same framework can reduce close-analogue contamination. We varied power over $\{2, 3, 4\}$ and grouping threshold over $\{0.3, 0.4, 0.5, 0.6\}$ on six full datasets, using exact fold counts and three seeds. Plain, power-only and group-only

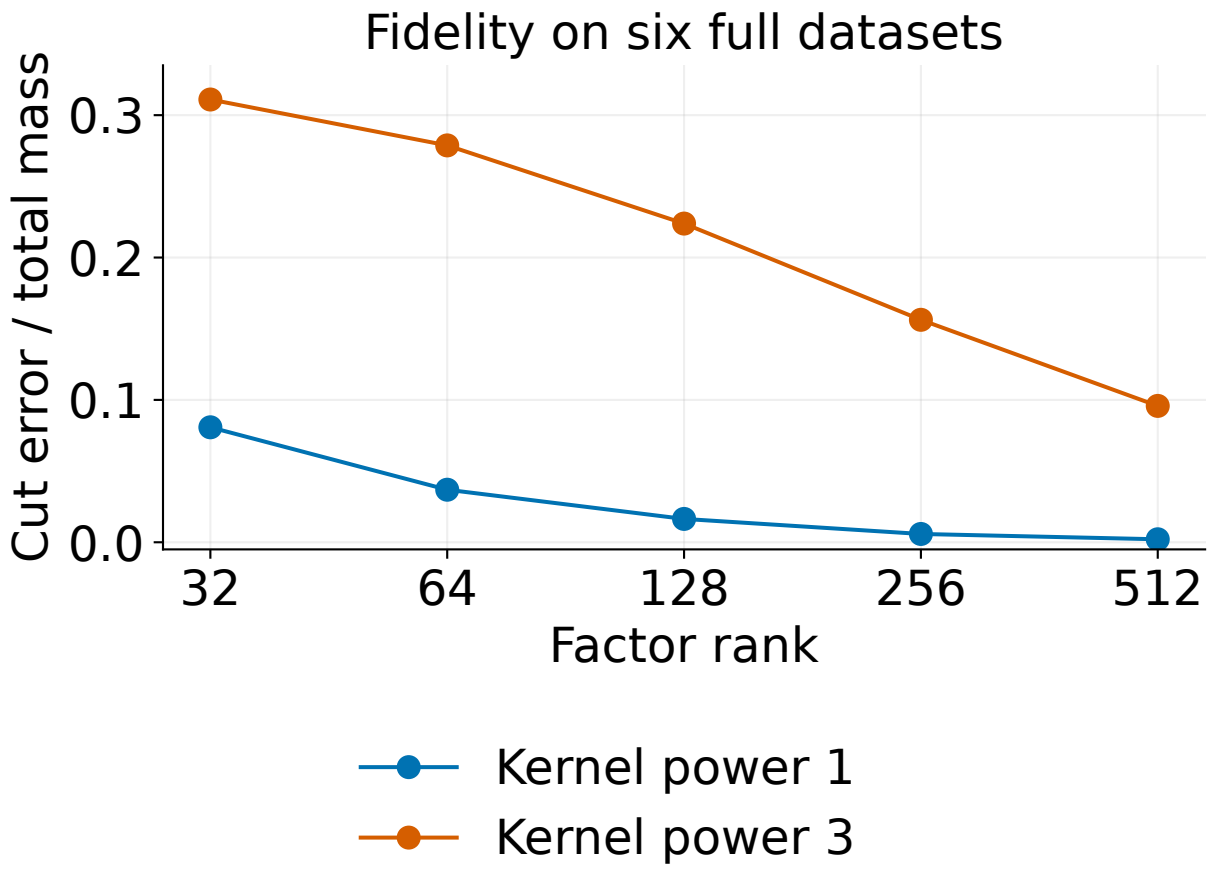



Figure 2: Cut-value fidelity for the ordinary and cubed kernels. Each point is the largest absolute factor-versus-exact cut difference, divided by the exact kernel's off-diagonal similarity mass, across six full datasets, three seeds and five partition perturbations. The powers have different denominators. The cubed kernel requires substantially more rank.

controls complete the 306-partition study. The resulting global–local frontier is supplied in the supplement. At the fixed combined setting, local leakage is lower than plain low rank on all six datasets under each of three Morgan ramps, the top-five-neighbour ramp and the fraction of test molecules above similarity 0.6. The improvement therefore extends beyond the ramp whose floor coincides with the grouping threshold. MACCS, used only for evaluation, shows lower mean nearest-training similarity on 4 of six datasets. Its dataset-bootstrap interval for the mean paired change includes zero. Local control is consequently strongest in the representation used to construct the split.

That control has a computational trade-off. For the cubed kernel, the maximum rank-256 cut error is 0.156 of powered mass, falling to 0.096 at rank 512 (Figure 2). The ordinary kernel's fidelity therefore cannot be assumed after sharpening. The normalized spectral estimates are often above one and give a vacuous uniform bound. Nevertheless, combined-setting local leakage changes less with rank: BACE's mean local ratio falls from 0.094 at rank 32 to 0.024 at rank 256 and 0.023 at rank 512. Grouping can preserve local structure

even when the powered factor is coarse. We therefore evaluate the returned partition as well as the factor's objective error. Full rank curves and alternative metrics appear in the supplement; accompanying result records retain group-relaxation counts.

## 3.4 Full-size baseline comparison

The full benchmark separates the benefit of compressing a dense objective from the effect of changing that objective. At tolerance 0.01, plain low rank has lower global overlap than the quality-preset graph comparator on 11 of 11 molecular datasets (Figure 3). The combined setting reduces local leakage relative to plain low rank on 11 of 11 datasets, while raising the ordinary dense cut. Its local score is lower than the graph comparator on 7 of 11 datasets (Figure 4). The mean combined-minus-graph local change is +0.007, reflecting the differing dataset effects. These comparisons expose the choice the framework offers: the plain factor directly targets distributed overlap; power and grouping spend some of that global advantage on local separation. The accompanying machine-readable tables include both tolerances, all reference methods, realized counts and group relaxations. Paired dataset effects, bootstrap intervals and Holm-adjusted sign tests are supplied in the supplement.

DataSAIL provides a closer comparison for the dense objective. Its saved assignments cover seven datasets, with source outcomes classified separately from missing attempts and resource-policy exclusions. At their exact native counts, plain low rank has lower global overlap on 7 of 7 datasets. The combined setting has lower global overlap on 4 of 7 and lower local leakage on 4, with 1 tie. Thus compression provides a practical dense-cut objective, while the local setting remains a choice among competing criteria. Reused assignments are rescored with the common protocol and supply no new runtime.

The published local and cluster methods clarify the retention contract. Lo-Hi satisfies its 0.4 cross-fold threshold on all seven eligible datasets, retaining between 98.0% and 99.9% of their molecules. Low-rank controls match each retained subset and its exact counts; their capped groups offer soft local control rather than that hard threshold guarantee. In the transferred UMAP recipe, seven native cluster holdouts and their count-matched controls retain every molecule. The supplement reports both studies separately. The FreeSolv native UMAP failure and MUV memory-policy exclusion remain unscored.

## 3.5 Dense splitting has a distinct computational advantage

The same-host comparison measures the public split call at exactly the native DataSAIL counts. Across four cases, median combined-setting times range from 0.56 to 2.70 seconds; DataSAIL takes 1,411 to 8,797 seconds, including distance construction, clustering and its integer program (Figure 5). Local quality remains dataset-dependent: on BACE the combined mean local ratio is 0.066, compared with 0.019 for DataSAIL. Speed and the desired separation criterion must therefore be considered together. Solver outcomes, initialization policy, individual timings and all exact counts appear in the supplement.

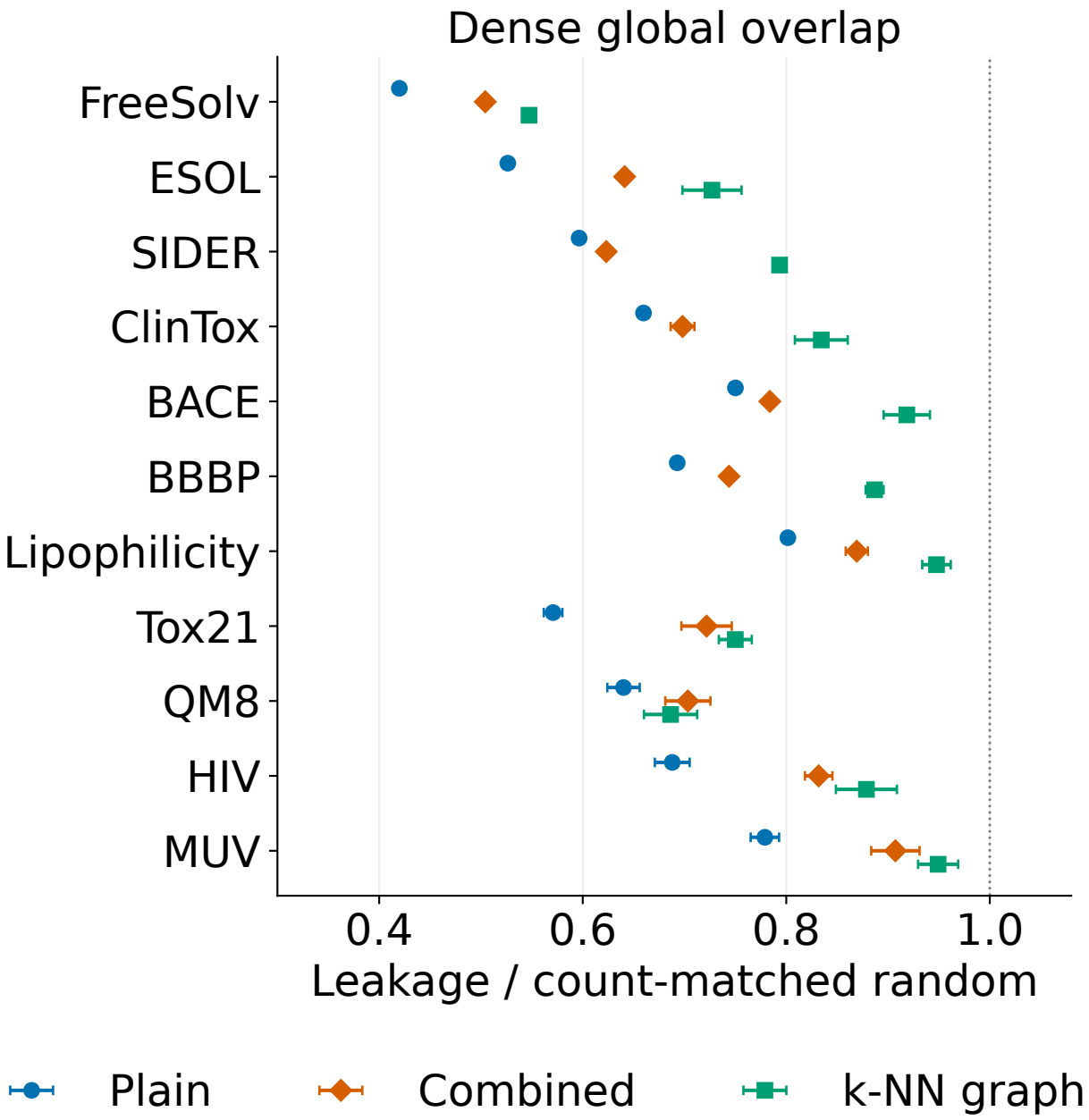



Figure 3: Global overlap on eleven full molecular datasets at size tolerance 0.01, ordered by dataset size. Points are means of three split seeds; bars show their standard deviations. The dotted line is the count-matched random expectation. All methods retain the same input entities. The graph comparator uses exact top-32 construction, an ordinary pairwise Graph API, the quality preset and CUT objective, with lower/upper count repair. The supplement reports all methods, actual counts, degree sensitivity and the two materials checks.

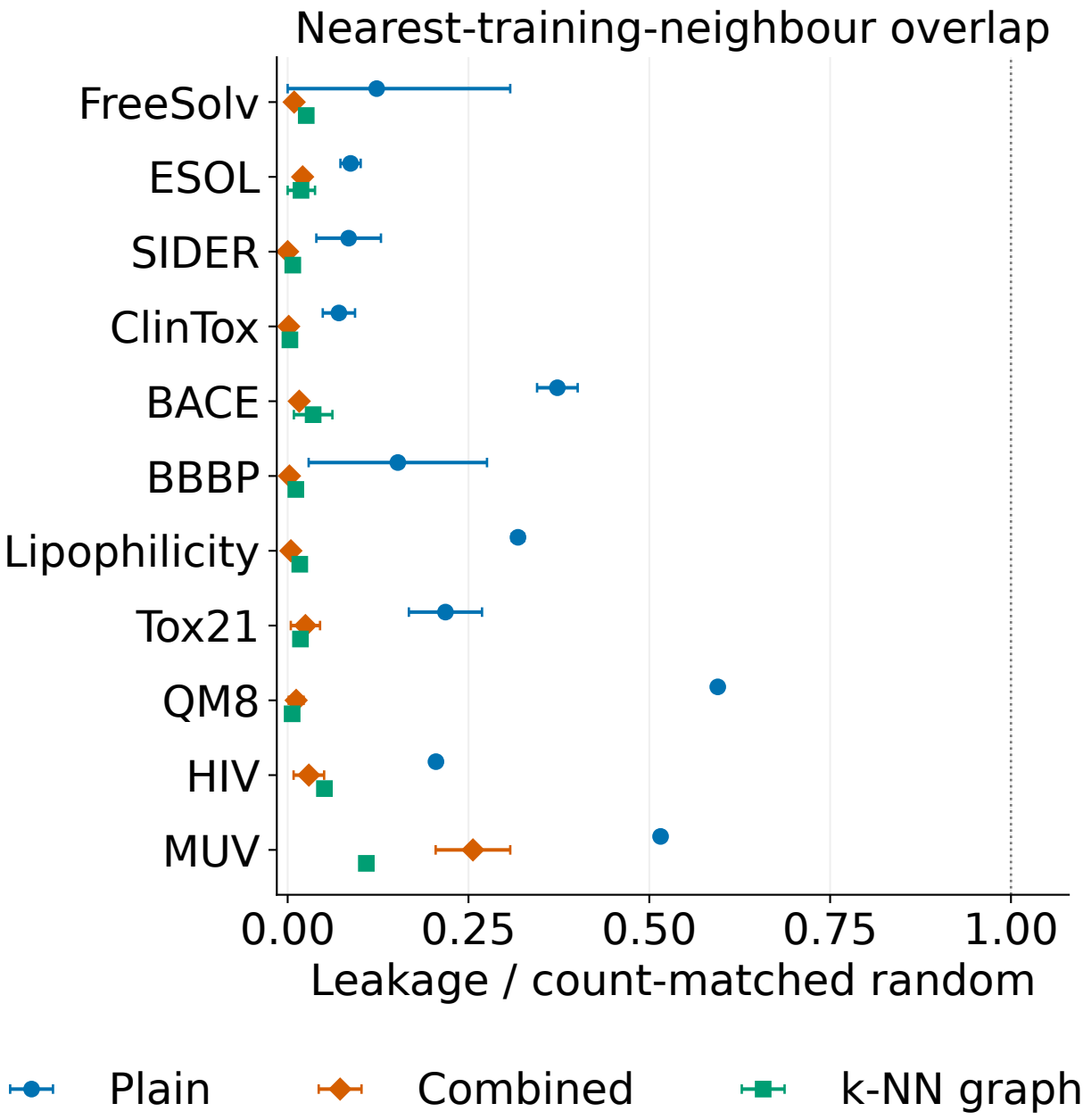



Figure 4: Test-only nearest-training-neighbour leakage for the same partitions and scoring protocol as Figure 3. Ratios are normalized by count-preserving random permutations. Bars show split-seed standard deviations, truncated at zero where necessary. Controlling total overlap and close-neighbour contamination gives different method orderings.

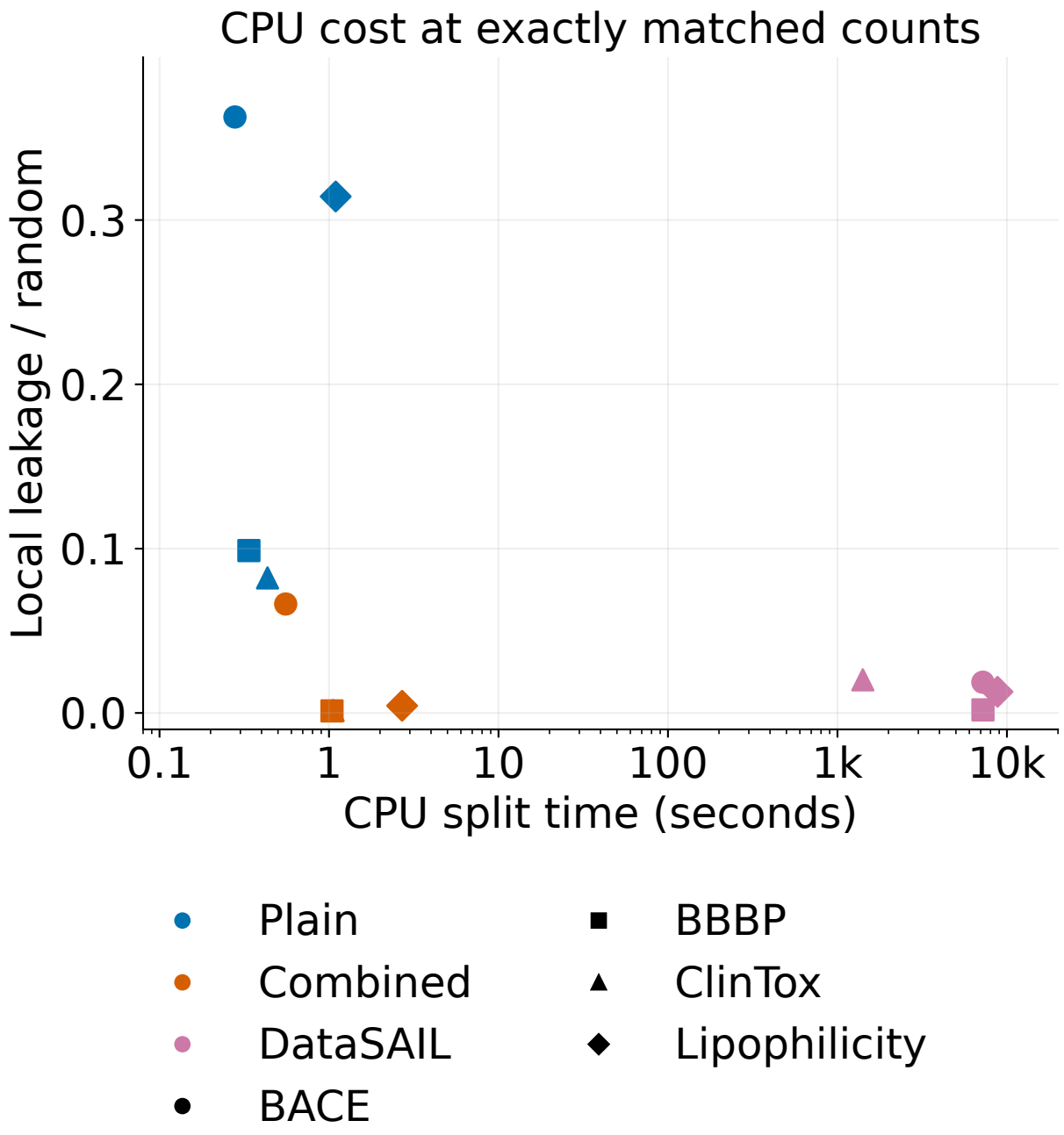



Figure 5: CPU cost and local overlap on four same-host cases at exactly matched DataSAIL fold counts. Colour denotes method and shape denotes dataset. Low-rank points average three split seeds for local leakage and use median split-call time; DataSAIL has one solve per case. The clock excludes feature loading and scoring. DataSAIL's distance construction, clustering and solver are included; low-rank calls exclude a 100-entity warm-up. Solver-reported optima and time-limited feasible outcomes are distinguished in the supplement.

### 3.6 Computational scaling differs between settings

On an H100 NVL, plain low rank partitions two million 2048-bit ChEMBL fingerprints in 19.9 seconds at size tolerance 0.01, with 25.3 GiB of peak GPU allocation. The measurement includes the public split call's factor construction, four restarts and refinement, after a small warm-up; loading features and evaluating leakage are outside the clock. The combined setting takes 89.0 seconds at 100,000 entities using exact tiled neighbours. These settings have different computational profiles (Figure 6). The combined pipeline is measured through 100,000 entities, and its performance is not extrapolated from the two-million-entity plain-factor result. Per-case memory, settings and output counts are supplied in the supplement.

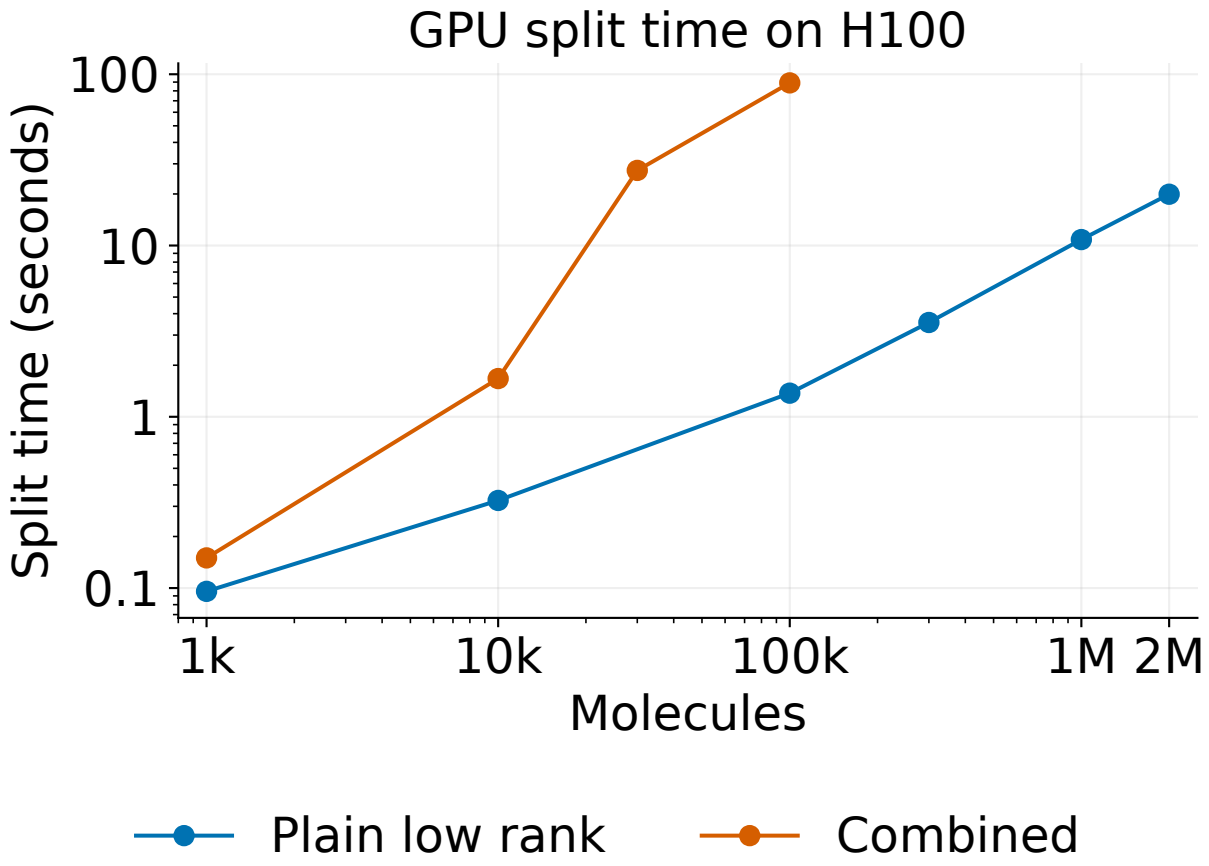



Figure 6: Split-call time on nested ChEMBL samples using the standard solver, rank 256 and size tolerance 0.01. All cases use H100 NVL hardware with two pinned CPU cores per worker. Each point is a fresh process with a 100-entity warm-up; feature loading and quadratic diagnostics are excluded. The plain and combined settings have separate measured ranges. k denotes thousands and M millions. Full per-case timings, GPU peaks and host RSS appear in the supplement.

### 3.7 Small-assay separation is strongly sensitive to target composition

On all ten prespecified external assays, the combined setting yields mean test ROC-AUC 0.522 for RF and 0.490 for boosted trees. Random splitting gives 0.871 and 0.863, respectively; the published SIMPD references give 0.867 and 0.853. In this assay collection SIMPD's aggregate scores are close to random, so a large drop from SIMPD is not by itself evidence of realistic project transfer. Several combined partitions produce below-chance AUC. Absolute scores and fold class proportions are supplied in the supplement.

At unchanged fold counts, matching class proportions raises the combined setting's mean AUC to 0.757 for RF and 0.725 for boosted trees (Figure 7). The intervention changes a median 6.6% of the combined assignments. Mean standardized target shift falls from 0.372 to 0.010. Plain low rank is also sensitive, changing from 0.589 to 0.765 for RF. The intervention simultaneously changes which chemical series appear in each fold, so it is not a causal estimate of the contribution of class proportions alone. Nevertheless, the original chance-level results cannot be interpreted simply as successful construction of a more useful benchmark.

These assays delimit the purpose of stronger structural separation. A new-series test can also separate the activity distributions associated with those series, as anticipated by the distinction between the Lo and Hi tasks [Steshin, 2023]. The combined setting is useful when that transfer is the intended question and sufficient predictive signal remains. For within-series interpolation, its imposed boundary can be excessive. Structural leakage, response composition and absolute model scores are needed together to assess the resulting benchmark.

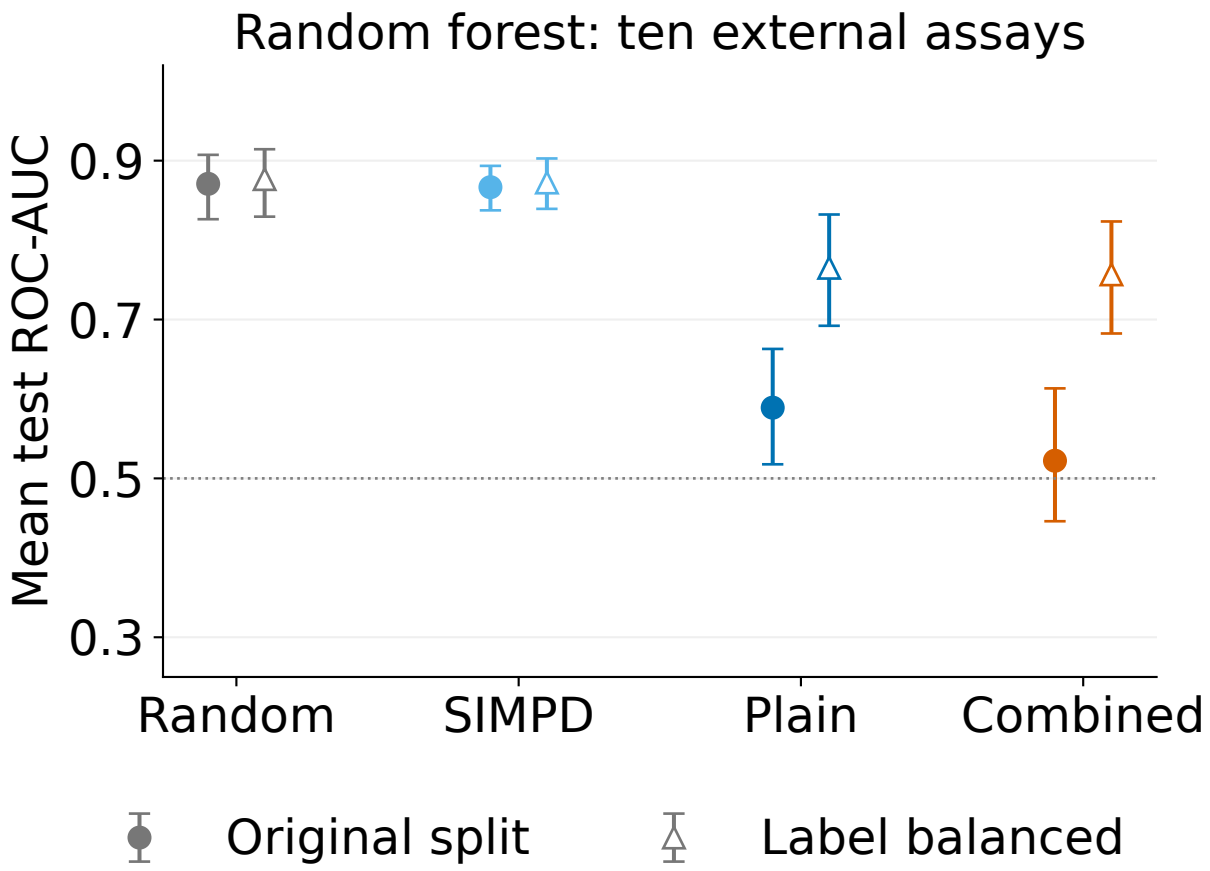



Figure 7: Absolute RF test AUC on ten external assays before and after the label-balance sensitivity intervention. Each assay is weighted equally after averaging split and model seeds. Error bars are descriptive 95% bootstrap intervals over assay means. The intervention uses target labels, preserves fold counts and can break structural groups. It is not part of the label-blind method. SIMPD supplies one published split per assay. The dotted line is AUC 0.5. Boosted-tree results and class proportions appear in the supplement.

## 4 Discussion

Low rank makes dense similarity an accessible objective for split construction. The factor-space identity replaces a stored pairwise matrix with shared coordinates and fold sums, and capacities provide direct control over the returned counts. This combination is useful for repeatedly constructing benchmarks, comparing overlap at matched sizes, and exploring how an evaluation boundary changes model performance. The measured fidelity of the ordinary kernel and the two-million-entity plain-factor run support this computational role. Its value depends on the kernel's compressibility and the chosen optimization settings.

The distinction between global and local similarity gives that computation a scientific interpretation. Total overlap measures a distributed relationship between folds; nearest-training-neighbour similarity measures opportunities for local analogue transfer. Powering and grouping let users change their relative importance, with an explicit cost in dense over-

lap, approximation requirements and neighbour construction. The Morgan sensitivity study supports local control beyond one threshold. The mixed independent-fingerprint results show why this control must be interpreted in the representation used to define similarity.

In practice, a user should first specify the intended transfer task and examine a plain partition at the required fold counts. The share of test molecules with a training neighbour above a chosen similarity threshold reveals whether local contamination remains relevant. Power and grouping can then strengthen separation for a new-series task. Fold composition, class proportions and absolute predictive scores should accompany that choice. The external assays show why maximizing separation alone is a poor selection rule: it can produce large target shifts and leave little learnable signal. The class-balance intervention changes structural composition as well as labels across folds, and therefore serves as a sensitivity check. Confirmatory benchmarks require the split policy to be fixed before the final model's test outcomes are viewed.

### 4.1 Limitations

Approximation, optimization and task validity impose separate limits. A small cut-value error does not certify a globally optimal partition. Kernel powering can require substantially more rank than ordinary similarity, while the spectral bound can be too loose to guide rank selection. Grouping adds exact neighbour work and only conditionally preserves groups under capacity repair. Consequently, the plain method's measured scaling cannot be transferred to the combined setting, and a hard analogue-exclusion requirement calls for a different retention or feasibility contract, as illustrated by Lo-Hi and GraphPart [Steshin, 2023, Teufel et al., 2023].

The empirical collection contains convenience datasets and single-target slices of multi-task sources, with incomplete DataSAIL coverage under the stated resource configuration. Composition descriptors define similarity in that representation and cannot establish crystal-structure novelty. The simulated SIMPD scenarios supply no observed future outcomes, and calibration depends on the selected labels and model. Broader tests should include chronological project data and additional representations, with independently specified deployment tasks. These limits guide where the framework can support benchmark design and where external evidence remains necessary.

## 5 Conclusion

Low-rank similarity splitting uses compressed fold sums to construct capacity-controlled partitions without storing the dense kernel. It supports a practical distinction between reducing total overlap and separating close analogues, with power and grouping controlling the trade-off. This makes low rank useful for designing and comparing scientific generalization tests, provided that structural separation is checked against the task, target composition and predictive behaviour.

# 6 Methods

## 6.1 Evaluation metrics

To account for realized fold sizes $n_c$, we also report the global ratio $L_S(z)/L_{\text{random}}$, where

$$L_{\text{random}} = 1 - \frac{\sum_c n_c(n_c - 1)}{n(n-1)}. \tag{4}$$

This expectation corresponds to random assignment at the same counts. It removes a direct size dependence of the global score, but does not remove differences in test composition or target distribution.

For local similarity, let $s_i = \max_{j:z_j=\text{train}} S_{ij}$ for each held-out entity. The nearest-neighbour score is the mean of $h(s_i) = \min\{1, \max\{0, (s_i - \ell)/(u - \ell)\}\}$. Molecular Tanimoto experiments use $(\ell, u) = (0.4, 0.8)$; these are operational thresholds for the chosen fingerprint, not universal definitions of chemical relatedness. For composition descriptors, the ramp uses the 0.99 and 0.999 quantiles of pair similarities estimated from at most 1024 sampled rows and 8192 sampled reference rows. We normalize by the mean score under random permutations of fold labels at fixed counts to obtain $\text{NNL}_{\text{rel}}$. A value near one denotes the random baseline, and zero denotes no sampled held-out neighbour above $\ell$; the relative score is not universally bounded by one. The main baseline comparison uses test-only queries and $k = 1$; validation-plus-test queries are a separate secondary measure of contamination relevant to model selection. The rank-sensitivity study retains that pooled held-out convention and labels it explicitly.

## 6.2 Datasets, baselines and prediction studies

The full-size benchmark covers eleven MoleculeNet datasets and two composition-based materials datasets, QMOF and a Materials Project snapshot [Rosen et al., 2021, Jain et al., 2013]. Molecular features are radius-two Morgan fingerprints with 1024 bits and Tanimoto similarity [Rogers and Hahn, 2010, Ralaivola et al., 2005]. Materials features combine MAGPIE-style elemental statistics, electronic, bonding and stoichiometric descriptors with a standardized Euclidean similarity $1/(1 + \|x_i - x_j\|)$ [Ward et al., 2016]. All feature construction and retained row identities are recorded. For multitask sources, the study uses one declared target, rather than claiming an evaluation across all tasks. The materials datasets provide a representation-specific generality check, reported in the supplement.

Comparator choice follows the question each method can answer. DataSAIL is the closest published comparator for dense cross-fold similarity. To isolate compression from sparsification, we use an ordinary weighted nearest-neighbour graph partitioned with Mt-KaHyPar's quality preset and edge-cut objective [Karypis and Kumar, 1998, Gottesbüren et al., 2024]. Graph construction is included in its split time. Lo-Hi's Hi splitter tests threshold-based chemical separation, allowing bridging molecules to be discarded; its comparisons match both the retained subset and realized counts [Steshin, 2023]. Published UMAP cluster splitting provides an embedding-and-clustering alternative [Hernández-Hernández and Ballester, 2023, Guo et al., 2025]. We transfer the former study's UMAP/Ward recipe, with every native cluster holdout receiving count-matched controls. The supplement gives the exact recipe

and eligibility policy. Random, scaffold and Butina splits supply familiar reference points. This focus on specified overlap levels is also consistent with SPECTRA's generalizability evaluation perspective [Ektefaie et al., 2024].

The full-size comparison uses an 8:1:1 ratio and size tolerances 0.01 and 0.05. Three split seeds quantify variation for plain low rank and the combined settings. DataSAIL 1.3.0 uses its C1e technique with SCIP, up to 7,200 seconds, cluster count $\min\{\max(9, \lfloor n/20 \rfloor), n\}$, and the stated size tolerance. DataSAIL has one recorded assignment per configuration. Solver logs distinguish time-limited feasible assignments, optima, solver failures, memory-policy skips and unattempted datasets. Local resource limits do not establish that another method cannot process a dataset in general.

Exact-size ablations use the same ratios with zero tolerance. Additional baselines assign entire Bemis–Murcko scaffolds or Butina clusters, using fixed fingerprints and a 0.6 distance cutoff for Butina [Bemis and Murcko, 1996, Butina, 1999]. Their indivisibility can cause count deviations, which we report rather than silently discarding examples. Butina's all-pairs implementation is limited to 10,000 entities. These baselines are informative structural comparators but are not exact-count controls when their realized counts differ.

The external assay study uses random forests and histogram-based gradient-boosted trees [Breiman, 2001, Chen and Guestrin, 2016] using 128 trees, two model seeds and three split seeds. Boosted trees use maximum depth four and learning rate 0.05. Settings are fixed across methods; no test-label tuning or per-split model selection is performed. Classification reports ROC-AUC and average precision; single-class test folds have undefined AUC and are not replaced with accuracy. Regression reports RMSE and $R^2$. Train–test target shift and classification rates accompany the scores. Model-initialization variability and split variability are retained separately.

Local-metric validation uses saved directed-message-passing neural-network predictions for random and plain-low-rank partitions of ESOL, FreeSolv, Lipophilicity and BBBP [Yang et al., 2019]. Their test counts match the native DataSAIL counts. Each partition uses a 30-epoch protocol, normalized RDKit 2D features and five model initializations, with one split realization and a random validation carve-out. Prediction and assignment provenance are specified in the supplement.

For external evaluation, ten of the 99 public SIMPD assay files are selected by sorting SHA256 hashes of filenames and taking the first ten, before inspecting their prediction outcomes [Landrum et al., 2023]. No parameters are selected on these assays. Their published simulated temporal split provides a scenario reference; new splits match its train/test counts exactly where the method permits. We use the provided activity labels with the same two tree-model families and retain all valid molecules. This evaluates transfer to assay tasks outside the development suite, not independence of every molecular structure or validation against observed future experiments.

## 6.3 Ablation, label balance and difficulty calibration

Six full-size development datasets support a component ablation separating kernel power, grouping, rank, landmarks, restarts, refinement, capacity tolerance, threshold and cap policy. We show the complete tested frontier rather than choose a winner using downstream test scores. These datasets also informed method development; the external assay study tests

transfer of the fixed settings.

A label-balance sensitivity analysis reassigns the minimum possible number of entities needed to match class proportions or five regression quantile bins across folds, while preserving fold counts. Rounding is solved as a small transportation problem. For that fixed integer target table, as many original assignments as possible are retained. Choices among interchangeable entities use a seeded shuffle. Both leakage measures and model scores are then recomputed. This intervention explicitly uses labels and can break groups; it diagnoses sensitivity to target distribution and is not a label-blind splitter or a causal isolation of structural novelty.

An optional hardness parameter uses swaps to randomize an optimized split while preserving fold counts. We assess difficulty relative to the random endpoint: the decrease in test AUC for classification, and the proportional increase in test RMSE for regression. RF calibration uses split seeds 0–2; seeds 10–14 test fixed knots and three interior targets, using RF and boosted trees. Labels are part of benchmark design in this study. The supplement reports both observed monotonicity and model-transfer error.

### 6.4 Scoring, uncertainty and reproducibility

The full-size baseline comparison uses one scoring protocol for every method. Global scores use all pairs up to 4,000 entities and a uniform 4,000-entity subset above that limit. Nearest-neighbour scoring uses all held-out entities up to 2,048 and a uniform subset above that limit, always against the complete training fold. Standardization is fitted to the complete feature set before similarity sampling. Three count-preserving permutations normalize nearest-neighbour scores. Sampling limits and reference counts accompany each result; sampled large-data global scores are not labelled exact. The powered-kernel fidelity study instead evaluates exact cuts of the full matrices on six development datasets with at most 5,000 entities. External assays are small enough for exact global and local scoring. These protocols answer different questions and their scope accompanies each table or figure.

Cross-dataset summaries average within each dataset first. Descriptive bootstrap intervals resample dataset means, with 10,000 replicates and seed 31415, avoiding molecule-level pseudo-replication. They describe the empirical collection and do not imply that thirteen convenience datasets are a random population sample. Per-seed tables and exceptions remain available. Supplementary uncertainty tables separate between-split variation from within-split initializer variation and use a split-then-model bootstrap with 2,000 draws and seed 2718. Single-split references cannot estimate split-choice variation. Split timing excludes separately computed diagnostics. Hardware, thread limits, source identities, input hashes, assignments and failures are retained in the accompanying manifests. The supplement identifies reused prediction studies and their source snapshots.

## Data and code availability

Code, manuscript sources, compiled PDFs and reproducibility material are available in the versioned repository release lowrank-v0.1.0-review-20261021-r2. The package includes source identities, environment versions, study settings, per-seed metrics, split assignments, input

checksums and scripts that rebuild the tables and figures. Download manifests identify the public SIMPD assays and original benchmark snapshots. MoleculeNet, QMOF, Materials Project and ChEMBL inputs retain their original source attribution and terms.

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## References

Zhenqin Wu, Bharath Ramsundar, Evan N. Feinberg, Joseph Gomes, Caleb Geniesse, Aneesh S. Pappu, Karl Leswing, and Vijay Pande. MoleculeNet: a benchmark for molecular machine learning. *Chemical Science*, 9(2):513–530, 2018. doi: 10.1039/C7SC02664A.

Kevin Yang, Kyle Swanson, Wengong Jin, Connor Coley, Philipp Eiden, Hua Gao, Angel Guzman-Perez, Timothy Hopper, Brian Kelley, Miriam Mathea, Andrew Palmer, Volker Settels, Tommi Jaakkola, Klavs Jensen, and Regina Barzilay. Analyzing Learned Molecular Representations for Property Prediction. *Journal of Chemical Information and Modeling*, 59(8):3370–3388, 2019. doi: 10.1021/acs.jcim.9b00237.

Simon Steshin. Lo-Hi: Practical ML Drug Discovery Benchmark. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/cb82f1f97ad0ca1d92df852a44a3bd73-Abstract-Datasets_and_Benchmarks.html.

Robert P. Sheridan. Time-Split Cross-Validation as a Method for Estimating the Goodness of Prospective Prediction. *Journal of Chemical Information and Modeling*, 53(4):783–790, 2013. doi: 10.1021/ci400084k.

Gregory A. Landrum, Maximilian Beckers, Jessica Lanini, Nadine Schneider, Nikolaus Stiefl, and Sereina Riniker. SIMPD: an algorithm for generating simulated time splits for validating machine learning approaches. *Journal of Cheminformatics*, 15:119, 2023. doi: 10.1186/s13321-023-00787-9.

Andreas Bender, Nadine Schneider, Marwin Segler, W. Patrick Walters, Ola Engkvist, and Tiago Rodrigues. Evaluation guidelines for machine learning tools in the chemical sciences. *Nature Reviews Chemistry*, 6(6):428–442, 2022. doi: 10.1038/s41570-022-00391-9.

Derek van Tilborg, Alisa Alenicheva, and Francesca Grisoni. Exposing the Limitations of Molecular Machine Learning with Activity Cliffs. *Journal of Chemical Information and Modeling*, 62(23):5938–5951, 2022. doi: 10.1021/acs.jcim.2c01073.

Roman Joeres, David B. Blumenthal, and Olga V. Kalinina. Data splitting to avoid information leakage with DataSAIL. *Nature Communications*, 16:3337, 2025. doi: 10.1038/s41467-025-58606-8.

Roman Joeres, David B. Blumenthal, and Olga V. Kalinina. Addendum: Data splitting against information leakage with DataSAIL. *Nature Communications*, 17:1597, 2026. doi: 10.1038/s41467-025-67495-w.

Christopher K. I. Williams and Matthias Seeger. Using the Nyström Method to Speed Up Kernel Machines. In *Advances in Neural Information Processing Systems*, volume 13, 2000. URL https://proceedings.neurips.cc/paper/2000/hash/19de10adbaa1b2ee13f77f679fa1483a-Abstract.html.

Inderjit S. Dhillon, Yuqiang Guan, and Brian Kulis. Kernel k-means, Spectral Clustering and Normalized Cuts. In *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 551–556, 2004. doi: 10.1145/1014052.1014118.

Alex Gittens and Michael W. Mahoney. Revisiting the Nyström Method for Improved Large-scale Machine Learning. *Journal of Machine Learning Research*, 17(117):1–65, 2016. URL https://jmlr.org/papers/v17/gittens16a.html.

Shusen Wang, Alex Gittens, and Michael W. Mahoney. Scalable Kernel K-Means Clustering with Nyström Approximation: Relative-Error Bounds. *Journal of Machine Learning Research*, 20(12):1–49, 2019. URL https://jmlr.org/papers/v20/17-517.html.

Radha Chitta, Rong Jin, Timothy C. Havens, and Anil K. Jain. Approximate kernel k-means. In *Proceedings of the 17th ACM SIGKDD international conference on Knowledge discovery and data mining*, pages 895–903, 2011. doi: 10.1145/2020408.2020558.

C.M. Fiduccia and R.M. Mattheyses. A Linear-Time Heuristic for Improving Network Partitions. In *19th Design Automation Conference*, pages 175–181, 1982. doi: 10.1109/dac.1982.1585498.

B. W. Kernighan and S. Lin. An Efficient Heuristic Procedure for Partitioning Graphs. *Bell System Technical Journal*, 49(2):291–307, 1970. doi: 10.1002/j.1538-7305.1970.tb01770.x.

Felix Teufel, Magnús Halldór Gíslason, José Juan Almagro Armenteros, Alexander Rosenberg Johansen, Ole Winther, and Henrik Nielsen. GraphPart: homology partitioning for biological sequence analysis. *NAR Genomics and Bioinformatics*, 5(4):lqad088, 2023. doi: 10.1093/nargab/lqad088.

Andrew S. Rosen, Shaelyn M. Iyer, Debmalya Ray, Zhenpeng Yao, Alán Aspuru-Guzik, Laura Gagliardi, Justin M. Notestein, and Randall Q. Snurr. Machine learning the quantum-chemical properties of metal–organic frameworks for accelerated materials discovery. *Matter*, 4(5):1578–1597, 2021. doi: 10.1016/j.matt.2021.02.015.

Anubhav Jain, Shyue Ping Ong, Geoffroy Hautier, Wei Chen, William Davidson Richards, Stephen Dacek, Shreyas Cholia, Dan Gunter, David Skinner, Gerbrand Ceder, and Kristin A. Persson. Commentary: The Materials Project: A materials genome approach to accelerating materials innovation. *APL Materials*, 1(1):011002, 2013. doi: 10.1063/1.4812323.

David Rogers and Mathew Hahn. Extended-Connectivity Fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010. doi: 10.1021/ci100050t.

Liva Ralaivola, Sanjay J. Swamidass, Hiroto Saigo, and Pierre Baldi. Graph kernels for chemical informatics. *Neural Networks*, 18(8):1093–1110, 2005. doi: 10.1016/j.neunet.2005.07.009.

Logan Ward, Ankit Agrawal, Alok Choudhary, and Christopher Wolverton. A general-purpose machine learning framework for predicting properties of inorganic materials. *npj Computational Materials*, 2(1):16028, 2016. doi: 10.1038/npjcompumats.2016.28.

George Karypis and Vipin Kumar. A Fast and High Quality Multilevel Scheme for Partitioning Irregular Graphs. *SIAM Journal on Scientific Computing*, 20(1):359–392, 1998. doi: 10.1137/s1064827595287997.

Lars Gottesbüren, Tobias Heuer, Nikolai Maas, Peter Sanders, and Sebastian Schlag. Scalable High-Quality Hypergraph Partitioning. *ACM Transactions on Algorithms*, 20(1):1–54, 2024. doi: 10.1145/3626527.

Saiveth Hernández-Hernández and Pedro J. Ballester. On the Best Way to Cluster NCI-60 Molecules. *Biomolecules*, 13:498, 2023. doi: 10.3390/biom13030498.

Qianrong Guo, Saiveth Hernandez-Hernandez, and Pedro J. Ballester. UMAP-based clustering split for rigorous evaluation of AI models for virtual screening on cancer cell lines. *Journal of Cheminformatics*, 17(1):94, 2025. doi: 10.1186/s13321-025-01039-8.

Yasha Ektefaie, Andrew Shen, Daria Bykova, Maximillian G. Marin, Marinka Zitnik, and Maha Farhat. Evaluating generalizability of artificial intelligence models for molecular datasets. *Nature Machine Intelligence*, 6(12):1512–1524, 2024. doi: 10.1038/s42256-024-00931-6.

Guy W. Bemis and Mark A. Murcko. The Properties of Known Drugs. 1. Molecular Frameworks. *Journal of Medicinal Chemistry*, 39(15):2887–2893, 1996. doi: 10.1021/jm9602928.

Darko Butina. Unsupervised Data Base Clustering Based on Daylight's Fingerprint and Tanimoto Similarity: A Fast and Automated Way To Cluster Small and Large Data Sets. *Journal of Chemical Information and Computer Sciences*, 39(4):747–750, 1999. doi: 10.1021/ci9803381.

Leo Breiman. Random Forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/a:1010933404324.

Tianqi Chen and Carlos Guestrin. XGBoost. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016. doi: 10.1145/2939672.2939785.